

Applied A.I. Solutions

Foundations of Data Management

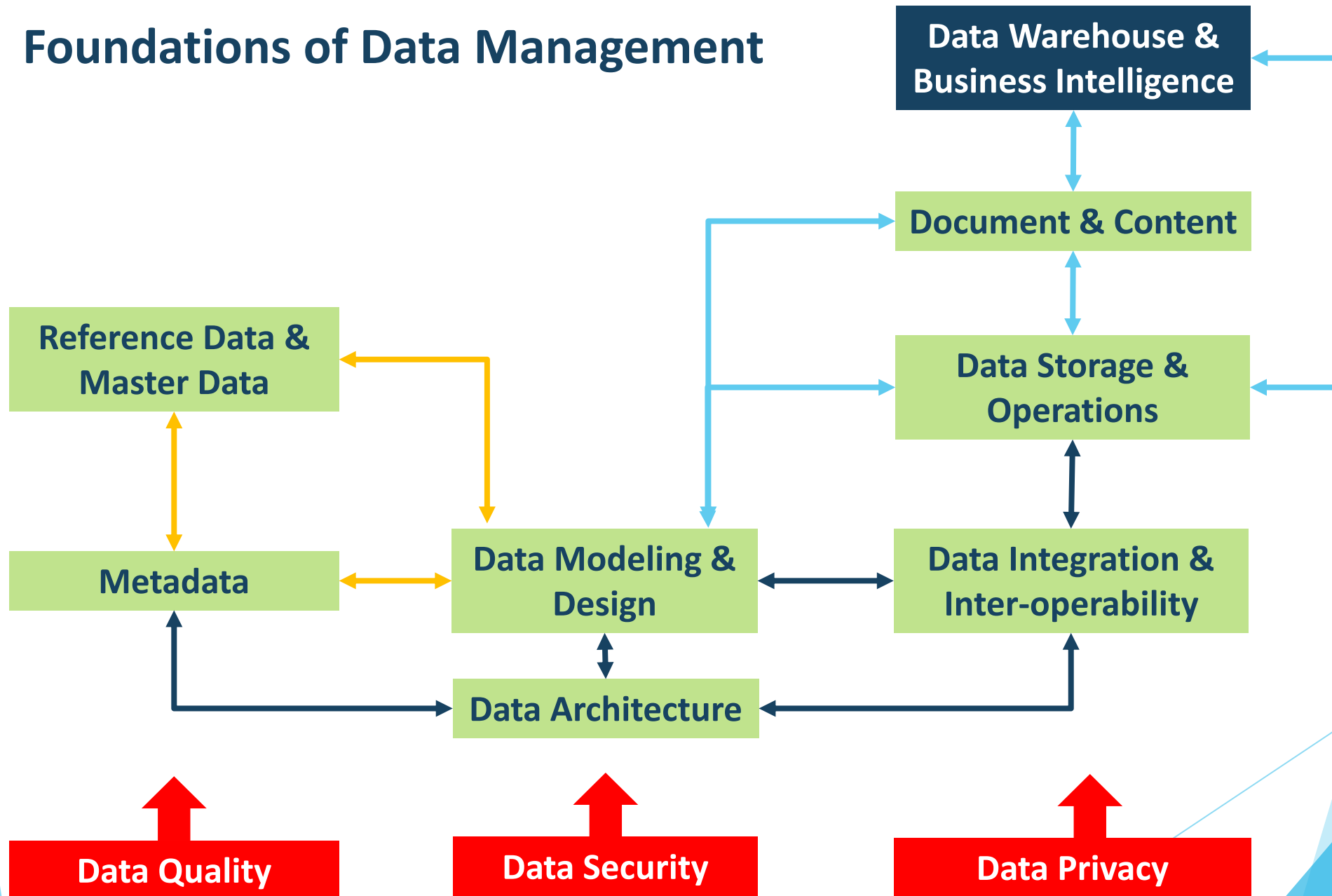
Professor

Daniel Vitaver, BSc (CSc), EMBA (Econ), PMP, CSM, CSSP

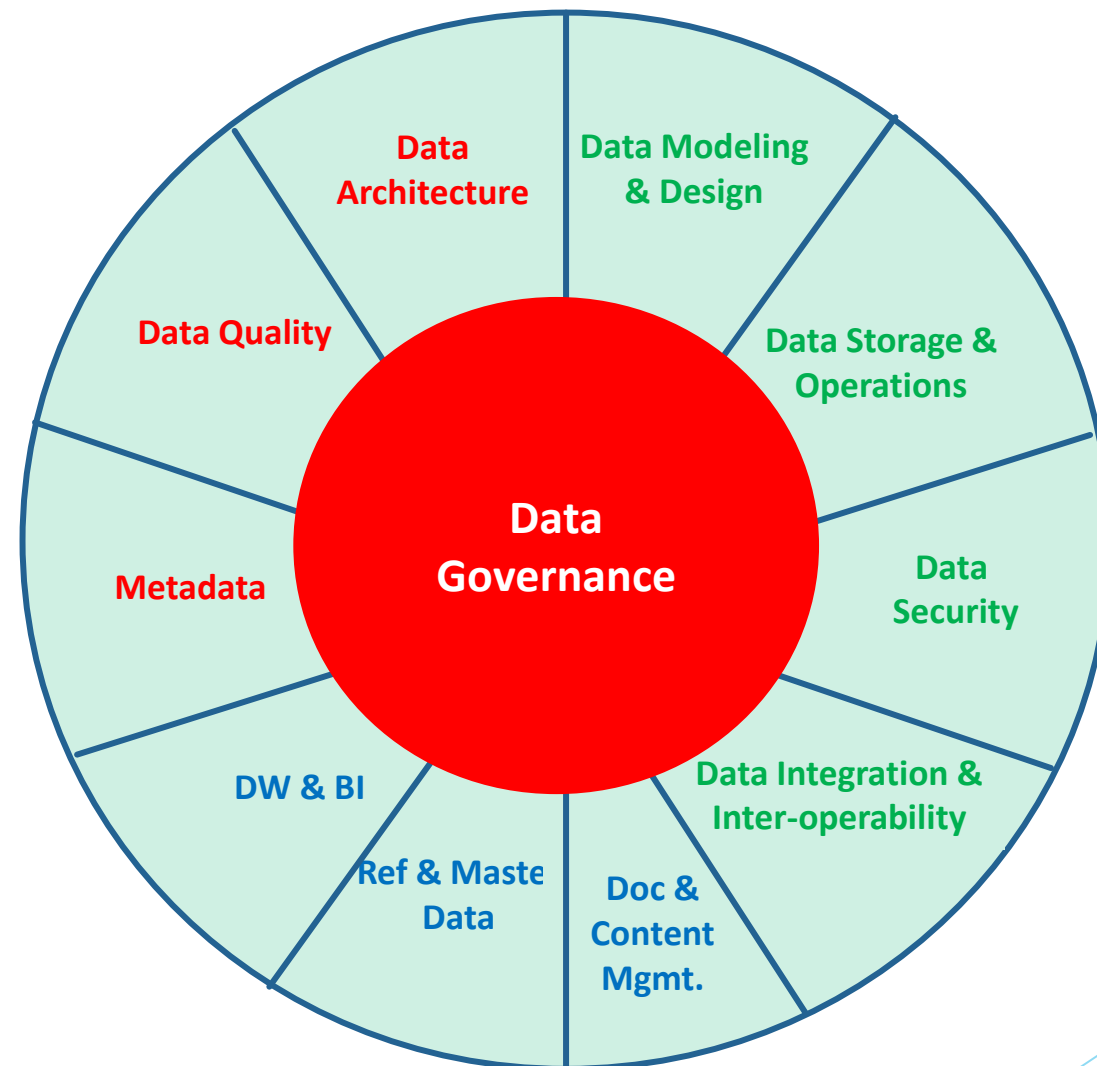
daniel.vitaver@georgebrown.ca

DATA GOVERNANCE

Foundations of Data Management

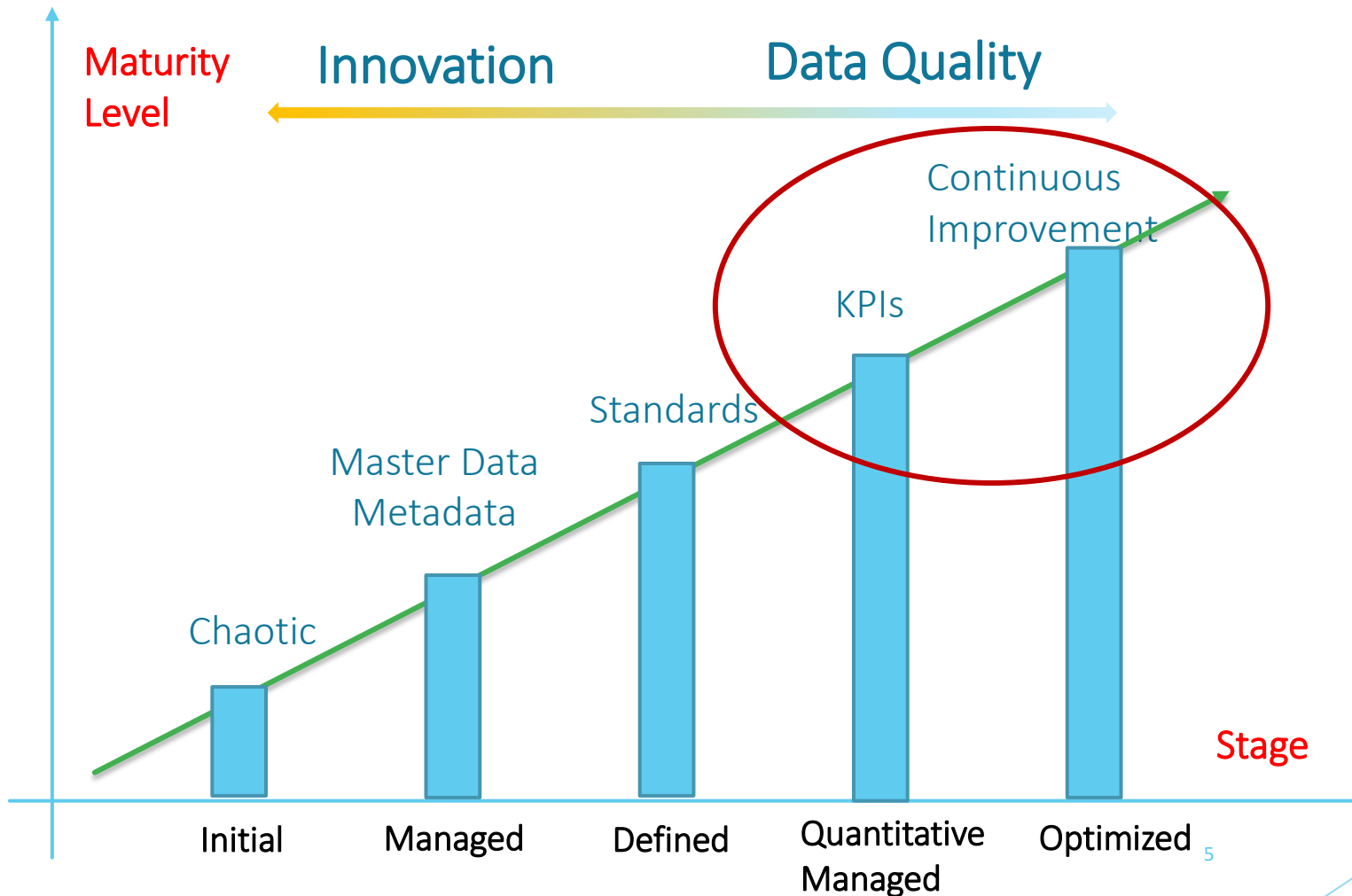


The DAMA Wheel



¹ Main source: Copyright © 2023 DAMA International – DMBOK2 - Technics Publications, Basking Ridge, New Jersey, USA

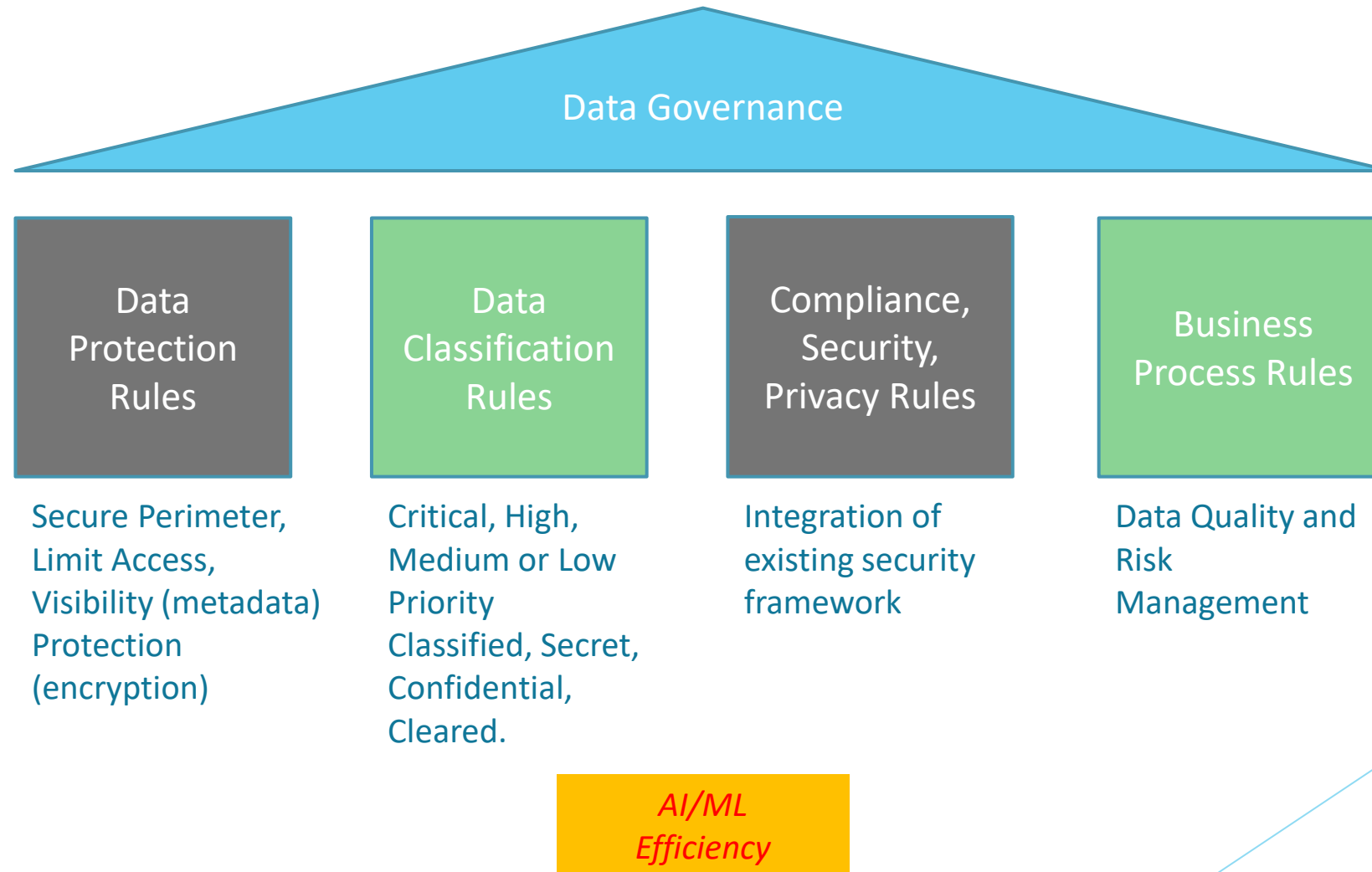
Data Management Maturity Model



Data Governance Program's Principles and Objectives

- Understanding the information **needs** of the business and its **stakeholders**
- Setting **High-Quality** Level standards for Data
- **Compliance**: ensuring, always, data-related regulatory compliance requirements
- **Data Asset Valuation**: setting standards to consistently define the value of data
- Ensuring **Privacy, Access and Confidentiality** of data

4-pillar Data Governance Framework



Data Governance Framework

Definition

The exercise of authority, control, and shared decision-making (planning, monitoring, and enforcement) over the management of data assets.

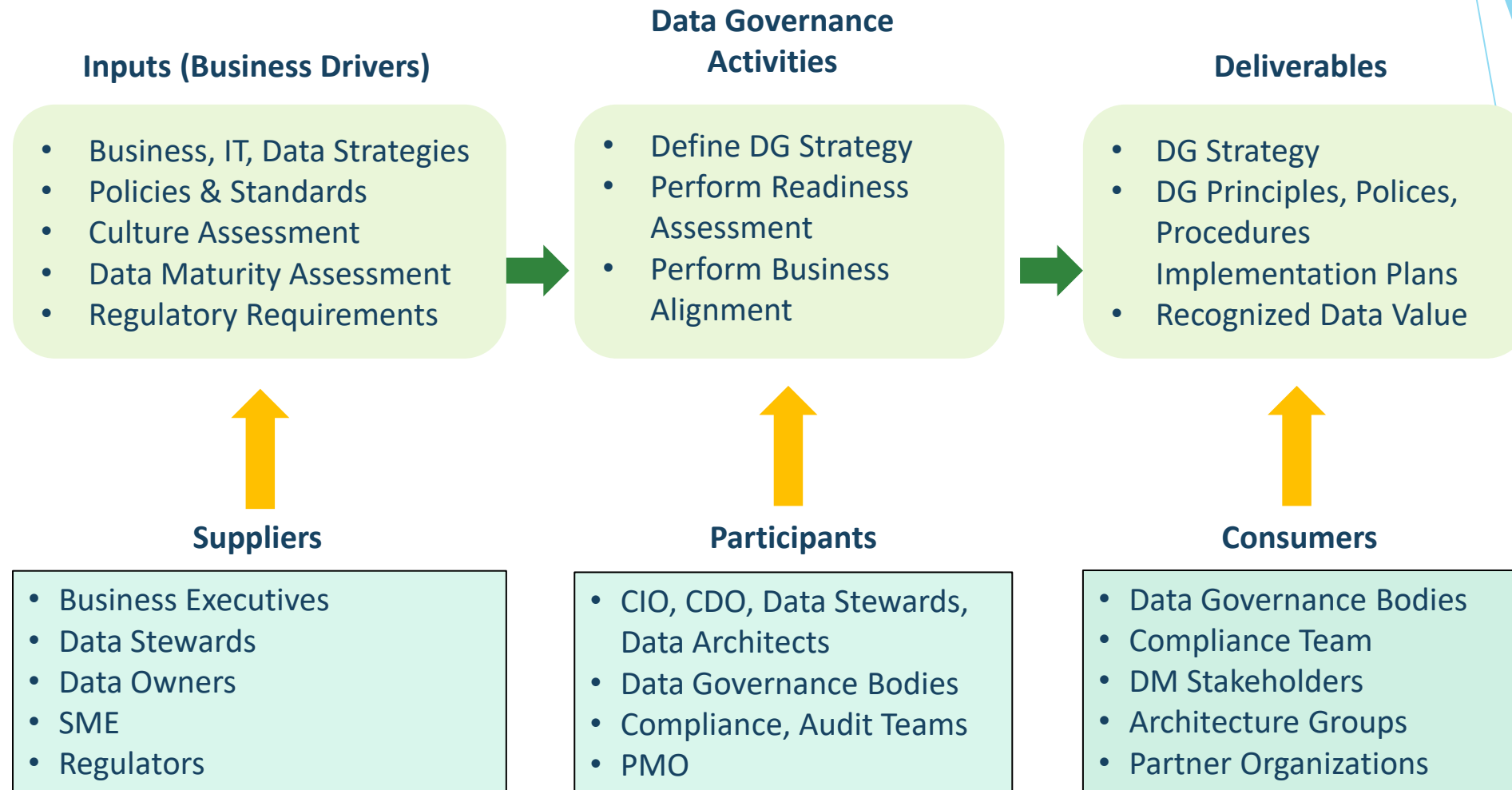
- ✓ who can take what actions
- ✓ with what information and when
- ✓ under what circumstances
- ✓ using what methods

Goals

1. Enable an organization to manage its data as an asset
2. Define, approve, communicate, and implement principles, policies, procedures, metrics, tools, responsibilities for DM
3. Monitor and guide policy compliance, data usage and management activities

DG drivers focus on

- Reducing risks: Risk management, Data Security, Privacy
- Improving processes: Regulatory Compliance, Data Quality, Metadata Management, Performance Management



Technical Drivers



Techniques

- Messaging
- Logo

Tools: software

- Business Glossary
- Workflow, Website, Scorecards
- Document Management

Performance Metrics

- Compliancy
- Value, Effectiveness, Sustainability

Data Stewardship

Label to describe **accountability and responsibility** for data and processes that ensure effective control and use of data assets.

Main activities

- Creating and Managing **Core Metadata**
- Documenting **rules and standards**
- Managing Data **Quality issues**
- Executing Operational **DG activities**

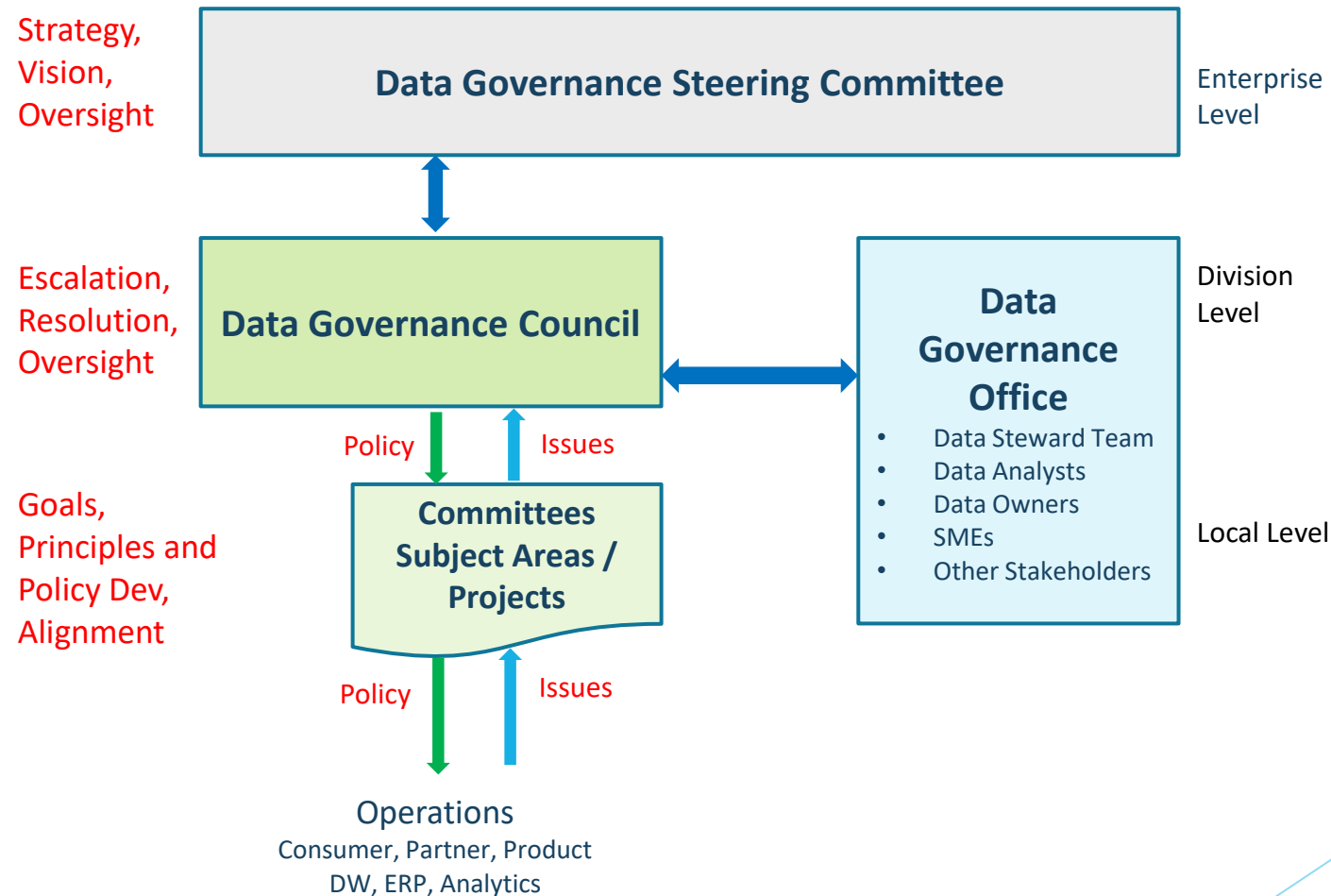
Types of Data Stewardship

A data steward **manages data assets** on behalf of others and in the best interest of the organization (McGilvray, 2008)

- Chief Data Steward/Officer
- Executive Data Stewards
- Enterprise Data Stewards
- Business Data Stewards
- Data Owner
- Technical Data Stewards
- Coordinating Data Stewards

¹ Source: Copyright © 2023 DAMA International – DMBOK2 - Technics Publications, Basking Ridge, New Jersey, USA

Data Governance Model



Data Policies

Data Policies are directives that **codify** principles and management intent into fundamental rules governing the...

- Creation
- Acquisition
- Integrity
- Security
- Quality
- Use of data and information

Metrics

a. **Value**

- Reduction of risk
- Contribution to business objectives
- Improved efficiency in operations

b. **Effectiveness**

- Extent stewards are using the relevant tools
- Achievement of Data Governance goals and objective
- Effectiveness of communication
- Effectiveness of education / training
- Speed of change adoption

c. **Sustainability**

- Conformance to regulatory requirements
- Performance of policies and process
- Conformance to standards and procedures

